

overall system is indigenous, while certain specialised electronic components are still imported due to limitations in India's current semiconductor ecosystem. However, we intend to collaborate with domestic semiconductor manufacturers as India's electronics and semiconductor industries continue to mature," he adds.

While discussing its revenue sources, Dr Kiran explains that the company generates revenue by carrying customers' payloads, sensors, communication equipment, and technology demonstrators on its VISTA near-space platform through both rideshare and dedicated missions.

The startup claims that its platforms can provide internet connectivity to underserved and remote regions where laying fibre-optic networks or installing cellular towers is economically challenging. A single near-space platform operating at an altitude of around 20 kilometres could potentially replace hundreds of traditional



Dr C.V.S. Kiran, Co-founder and CEO, Red Balloon Aerospace

cell towers while delivering high-speed connectivity. The company claims that its systems can achieve data speeds of up to 500Mbps with latency as low as 0.3 milliseconds, significantly outperforming many satellite-based solutions.

According to its estimates, the company's services can be significantly cheaper while offering up to 200 times

the operational efficiency in certain applications. Unlike satellites, which are often discarded after their operational life, Red Balloon's ALTIS and HELIX airships are reusable. Payloads can be safely recovered, upgraded, modified, and redeployed, reducing both costs and environmental impact.

While discussing future plans, Dr Kiran shares that the company plans to launch ALTIS and a subscale version of HELIX.

The company's name is inspired by the French Oscar-winning film *The Red Balloon*. Kiran explains that the persistence and determination shown by the young boy in the film reflect the company's philosophy of building platforms capable of persistent operation.

Currently, the company operates from a 20,000-square-foot (approx. 1858 sq. metres) facility in Vijayawada and employs around 24 employees, with plans to expand its workforce to nearly 150 in the coming year.

—Nitisha Dubey

2 SEGRITECH'S COMPACT AND PORTABLE AUTOMATED MACHINE SORTS FRUITS AND VEGETABLES AT FARM LEVEL

This startup has developed a computer vision-based machine that inspects fruits, grades and sorts them by quality, and identifies and separates defective produce directly on farms. Could this be a significant advancement in post-harvest processing?

The idea for the startup came from a problem Hetendra Singh Rathore, Vijay Pratap Singh, and Sri Kusulu Devalla faced while running their earlier supply chain business for fresh fruits and vegetables. Every fruit had to be sorted by hand, which took time, required significant manual labour, and made it difficult to handle large orders. After Covid-19 affected their business, they decided to solve this problem instead. They started working on a machine that could sort and grade fruits automatically and founded Segritech in 2022. After three years of research and development, they launched their



(R to L) Hetendra Singh Rathore, Founder, Sri Kusulu Devalla, Co-Founder and CTO, and Vijay Pratap Singh, Co-Founder, Segritech

machine to sort and grade fruits and vegetables at the farm level, helping farmers process their produce before it reaches the market.

Most sorting machines are designed for factories and are

expensive. Segritech built a compact and portable machine that can be moved from one farm to another. The machine measures about 10 feet by 5 feet (3m × 1.5m approx.) and is designed for farmers and local traders.